

## **TECH STACK:**

| <b>Category</b>            | <b>Technology</b>                                            | <b>Purpose</b>                                         |
|----------------------------|--------------------------------------------------------------|--------------------------------------------------------|
| <b>Frontend</b>            | React.js, Vite, HTML5, CSS3, JavaScript                      | Build the web application and user interface           |
| <b>UI Framework</b>        | Tailwind CSS                                                 | Responsive and modern UI design                        |
| <b>Maps &amp; Location</b> | Leaflet.js + OpenStreetMap                                   | Display problem locations and demand heatmaps          |
| <b>Backend</b>             | Node.js, Express.js                                          | Build REST APIs and handle business logic              |
| <b>Database</b>            | MongoDB, Mongoose                                            | Store users, reports, opportunities, and analytics     |
| <b>AI/ML</b>               | Python, Scikit-learn, Pandas, NumPy                          | Categorize reports, detect duplicates, estimate demand |
| <b>Authentication</b>      | JWT (JSON Web Token), bcrypt                                 | User authentication and password security              |
| <b>Image Storage</b>       | Cloudinary                                                   | Store uploaded images securely                         |
| <b>Version Control</b>     | Git, GitHub                                                  | Source code management and team collaboration          |
| <b>API Testing</b>         | Postman                                                      | Test backend APIs                                      |
| <b>Testing</b>             | Jest, Supertest                                              | Unit and API testing                                   |
| <b>Deployment</b>          | Vercel (Frontend), Render (Backend), MongoDB Atlas           | Host the application and database                      |
| <b>Documentation</b>       | Markdown (README), Google Docs, Canva / Microsoft PowerPoint | Project documentation and presentation                 |
| <b>Project Management</b>  | GitHub Projects, GitHub Issues                               | Track tasks and project progress                       |